1. Asking Meaningful Questions

Before starting the analysis, the following key questions were defined:

How do stock prices change over time?

What is the distribution of closing prices?

Is there a relationship between opening, closing, high, and low prices?

Are there any unusual spikes or drops in stock prices?

Which variables are most strongly correlated?

These questions guided the exploration and helped focus on extracting useful insights.

2. Exploring Data Structure

The dataset was examined to understand its structure:

Variables Identified:

Date (Datetime)

Open (Numeric)

High (Numeric)

Low (Numeric)

Close (Numeric)

Volume (Numeric)

Data Types:

Date → Converted to datetime format

Price-related columns → Float

Volume → Integer

Dataset Overview:

Contains daily stock price records

No major structural inconsistencies found

Summary Statistics:

Mean, median, min, max values calculated

Provided insights into price ranges and variability

3. Identifying Trends, Patterns, and Anomalies

📈 Trends

Stock prices show fluctuations over time

Some periods indicate steady growth, while others show decline

📊 Patterns

Strong relationship between:

Open and Close prices

High and Low prices

Prices generally move together, indicating consistency in market behavior

⚠️ Anomalies

Sudden spikes or drops in prices observed

These may be due to:

Market events

Economic news

External shocks

4. Testing Hypotheses and Validating Assumptions

Hypothesis 1:

☞ Opening price is strongly related to closing price

Tested using correlation analysis

Result: Strong positive correlation

✔ Hypothesis accepted

Hypothesis 2:

☞ High prices increase when closing prices increase

Verified using scatter plot & correlation

Result: Strong relationship observed

✔ Hypothesis accepted

Hypothesis 3:

☞ Stock prices follow a consistent trend over time

Tested using time series plot

Result: Prices fluctuate significantly

✕ Hypothesis partially rejected

5. Detecting Data Issues

During analysis, the following potential issues were checked:

🔍 Missing Values

No significant missing values found

🔁 Duplicate Records

No major duplicates detected

⚠ Outliers

Some extreme price values observed

These may represent real market behavior rather than errors

🔧 Data Type Issues

Date column required conversion (fixed)

Conclusion

The Exploratory Data Analysis revealed that:

Stock prices are highly correlated across variables

There are clear fluctuations over time rather than stable trends

The dataset is mostly clean and reliable

Some extreme values exist but are likely valid market movements

EDA helped in understanding the dataset structure, identifying patterns, and preparing the data for further advanced analysis or modeling.